

Bill of Materials

All parts are through-hole, 2.54mm pitch unless otherwise stated.

Links are provided for convenience and may not correspond exactly to the recommended part. Always verify dimensions and specifications against the datasheet before ordering.

Module	Part	Qty	Value	Note
All Modules	2.0mm Pitch 16pin Connector	1	Socket	Standard module connector
	Electrolytic Cap	1	10uf 16v	Main power decoupling
	Round Neodymium Magnets	4	5mm/5mm	Requires high precision
	PCB	1		
	Acrylic Face Plate	1		
	Screw	2	3M, 15mm	Telemetry needs 4
	Nut	4	3M	Telemetry needs 16
Abort	Circle Button	1	Red, 12mm	OFF-(ON)
	Safety Switch	1	Red	
	LED	1	Red, 10mm	
	Chip Socket	1	8 Pin	
	Resistor	1	1.2K, 1/4W	
	Resistor	1	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
Action Groups	Attiny25	1		
	Square Button	10	Blue	
	LED	10	Blue, 3mm	
	LED Socket	10	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	4	16 Pin	
	Resistor	10	10k, 1/4W	
	Resistor	10	62k, 1/4W	
	Ceramic Capacitor	6	100nf	Type 104
	Attiny45	2		
	74HC165	2		
Analog	74HC595	2		
	Square Button	2	Blue	
	Switch	2	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	7	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
Analog Throttle	74HC165	1		
	Square Button	2	Blue	
	Switch	1	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	6	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
	74HC165	1		
	Potentiometer	1	B20K	
Camera	Potentiometer Cover Dial	1		
	Square Button	4	Black	
	Square Button	6	White	
	Chip Socket	1	8 Pin	
	Chip Socket	2	16 Pin	
	Resistor	10	10k, 1/4W	
	Ceramic Capacitor	3	100nf	Type 104
	Attiny45	1		
	74HC165	2		

Module	Part	Qty	Value	Note
Control Systems	Square Button	10	White	
	Switch	2	SPST	
	LED	10	Orange, 3mm	
	LED Socket	10	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	4	16 Pin	
	Resistor	22	10k, 1/4W	
	Ceramic Capacitor	6	100nf	Type 104
	Attiny45	2		
	74HC165	2		
	74HC165	2		
Executive Actions (Control)	Square Button	1	White	
	Square Button	2	Blue	
	Circle Button	1	Green, 12mm	OFF-(ON)
	LED Button	1	Red, 16mm	
	Button Safety Cover	1	22x24mm	
	Switch	2	SPST	
	Key Switch	1	12mm	
	LED	3	Orange, 3mm	
	LED	2	Blue, 3mm	
	LED Socket	5	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	2	16 Pin	
	Resistor	1	1.2K, 1/4W	
	Resistor	10	10k, 1/4W	
	Resistor	2	62k, 1/4W	
	Ceramic Capacitor	4	100nf	Type 104
	Attiny45	2		
	74HC595	1		
	74HC165	1		
Executive Actions (Groups)	Square Button	5	Blue	
	Circle Button	1	Green, 12mm	OFF-(ON)
	LED Button	1	Red, 16mm	
	Button Safety Cover	1	22x24mm	
	Key Switch	1	12mm	
	LED	5	Blue, 3mm	
	LED Socket	5	3mm	
	Chip Socket	2	8 Pin	
	Chip Socket	2	16 Pin	
	Resistor	1	1.2K, 1/4W	
	Resistor	7	10k, 1/4W	
	Resistor	5	62k, 1/4W	
	Ceramic Capacitor	4	100nf	Type 104
	Attiny45	2		
	74HC595	1		
	74HC165	1		
Executive Actions	Circle Button	1	Green, 12mm	OFF-(ON)
	LED Button	1	Red, 16mm	
	Button Safety Cover	1	24x26mm	
	Key Switch	1	12mm	
	LED	1	Green, 5mm	
	LED Socket	1	5mm	
	Chip Socket	1	8 Pin	
	Resistor	1	300, 1/4W	
	Resistor	1	1.2K, 1/4W	
	Resistor	2	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny25	1		

Module	Part	Qty	Value	Note
EVA	Square Button	6	White	
	Square Button	5	Green	
	LED Array	1	Green, 10 LED	
	Chip Socket	2	8 Pin	
	Chip Socket	4	16 Pin	
	Resistor	10	1.2K, 1/4W	
	Resistor	11	10k, 1/4W	
	Ceramic Capacitor	6	100nf	Type 104
	Attiny45	2		
	74HC1595	2		
	74HC165	2		
Navigation	Square Button	6	Black	
	LED Switch	2	Yellow	
	Chip Socket	1	8 Pin	
	Chip Socket	1	16 Pin	
	Resistor	2	1.5k, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny45	1		
	74HC165	1		
Natigation (Time)	Square Button	2	Black	
	Square Button	2	Yellow	
	LED Button	1	Yellow, 16mm	
	D36 LED Switch	1	Yellow	
	6P Rocker Switch	1		
	Chip Socket	2	8 Pin	
	Chip Socket	1	16 Pin	
	Resistor	1	300, 1/4W	
	Resistor	1	1.5k, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	3	100nf	Type 104
	Attiny45	2		
	74HC165	1		
Rotation	Square Button	3	Green,	
	Switch	1	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	6	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
	74HC165	1		
Rotation (Throttle)	Square Button	2	Green,	
	Switch	1	SPST	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Chip Socket	1	16 Pin	
	Resistor	4	10k, 1/4W	
	Ceramic Capacitor	2	100nf	Type 104
	Attiny24	1		
	Potentiometer	1	B20K	
	Potentiometer Cover Dial	1		

Module	Part	Qty	Value	Note
Stage	Circle Button	1	Green, 12mm	OFF-(ON)
	Safety Switch	1	Green,	
	LED	1	Green, 10mm	
	Chip Socket	1	8 Pin	
	Resistor	1	150, 1/4W	
	Resistor	1	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny25	1		
Telemetry	Square Button	4	Black	
	LCD Screen	1	128x64	ST7920 driven
	Header Pins	1	1x20 Pin	Not socket
	Resistor	4	10k, 1/4W	
	Washer	8	3M	Depends on thickness
	Arduino Nano	1		
Throttle	Square Button	1	Yellow	
	Square Button	1	Red	
	Switch	1	SPDT 3-Pos	ON-OFF-ON
	Potentiometer	1	B20k, 6 Pin	Two parallel channels
	Throttle Lever	1		3D printed
	Throttle Friction Mechanism	1		3D printed
	Chip Socket	1	8 Pin	
	Resistor	2	10k, 1/4W	
	Diode	1	1N4148	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny25	1		
Time	Square Button	3	Black	
	Square Button	2	Yellow	
	LED Button	1	Yellow, 16mm	
	6P Rocker Switch	1		
	Chip Socket	2	8 Pin	
	Chip Socket	1	16 Pin	
	Resistor	1	300, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	3	100nf	Type 104
	Attiny45	2		
	74HC165	1		
Translation	Square Button	3	Green,	
	Switch	2	SPST	
	JH-D400X Joystick	1		
	Chip Socket	1	14 Pin	
	Resistor	5	10k, 1/4W	
	Ceramic Capacitor	1	100nf	Type 104
	Attiny24	1		
Utility (Navigation)	Square Button	2	Black	
	Square Button	5	Green	
	D36 LED Switch	1	Yellow	
	LED Array	1	Green, 10 LED	
	Chip Socket	2	8 Pin	
	Chip Socket	3	16 Pin	
	Resistor	10	1.2K, 1/4W	
	Resistor	1	1.5k, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	5	100nf	Type 104
	Attiny45	2		
	74HC595	2		
	74HC165	1		

Module	Part	Qty	Value	Note
Utility (Time)	Square Button	3	Black	
	Square Button	5	Green	
	LED Array	1	Green, 10 LED	
	Chip Socket	2	8 Pin	
	Chip Socket	3	16 Pin	
	Resistor	10	1.2K, 1/4W	
	Resistor	8	10k, 1/4W	
	Ceramic Capacitor	5	100nf	Type 104
	Attiny45	2		
	74HC595	2		
	74HC165	1		
Quad Hub	2.0mm Pitch 16pin Connector	4	Socket	Standard module connector
	Resistor	2	4.7k, 1/4W	
	Nut	6	3M	
	Arduino Micro	1		Can be substituted with a Nano
	Header Sockets	2	15 Pin, Socket	For the arduino to slot into
	Barrel Jack	1	2.1mm	
Octo Hub	2.0mm Pitch 16pin Connector	8	Socket	Standard module connector
	Resistor	2	4.7k, 1/4W	
	Nut	6	3M	
	Arduino Micro	1		Can be substituted with a Nano
	Header Sockets	2	15 Pin, Socket	For the arduino to slot into
	Barrel Jack	1	2.1mm	
Miscellaneous	Power Supply	1	5V	Center 5V, Sleeve ground
	Micro USB cable	1		
	Neodymium Magnets	10	5mm/5mm	
	Ribbon Cable	2-3m	16 wire, 1mm	Depending on size
	Ink	1	Black	Alcohol or acetone based
	Acrylic Container	1		Depending on size
	2.0mm Pitch 16pin Connector	2-16	Pin	Two per module

Required Tools

- Soldering iron and all relevant materials (solder, flux, etc.)
- Wire (standard hobby gauge, ~22-24 AWG)
- Wire strippers/cutters
- Pliers
- Screwdriver, with different size tips
- Vise/vice or some reliable way to compress objects
- 8mm wrench (for all switches and 3mm LEDs)
- 9mm wrench (Analog Throttle and Rotation Throttle modules)
- 15mm wrench (Executive Actions, Stage, Abort modules)
- Ink (for lettering, black or your choice)
- A suitable way to clean the ink (such as acetone-based nail polish remover)
- Silicone caulk (clear or the color of your container)
- A breadboard
- Arduino IDE software or similar
- Multimeter